

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME:CSA1610

COURSE CODE: Data Warehousing and Data Mining

Student Name:-D.Deepak

REG NO:-192572095

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A MOCK INTERVIEW

<i>Criteria</i>	Technical Knowledge <i>(2.5)</i>	Problem Solving <i>(2.5)</i>	Analytical Thinking <i>(1.5)</i>	<i>Communication(1.5)</i>	Confidence <i>(1)</i>	Response to Questions(1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	15%	15%	10%	10%	100%
<i>Q1</i>							
<i>Q2</i>							

Code of Conduct:

I D. Deepak/192572095 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student :deepak

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

Q1. Dataset: Retail Transactions

TID	Items
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question:

Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example

ANS:-

Association Rule Mining in Retail Transactions

Association Rule Mining is a data mining technique used to discover relationships between items that are frequently purchased together. In the given dataset, the items are Milk, Bread, Butter, and Eggs.

Step 1: Identify frequent combinations

From the transactions:

- Milk appears in T1, T2, T4, T5 $\rightarrow 4/5 = 80\%$ support
- Bread appears in T1, T3, T4 $\rightarrow 3/5 = 60\%$ support
- Butter appears in T2, T3, T4 $\rightarrow 3/5 = 60\%$ support
- Milk and Bread appear together in T1 and T4 $\rightarrow 2/5 = 40\%$ support
- Milk and Butter appear together in T2 and T4 $\rightarrow 2/5 = 40\%$ support
- Bread and Butter appear together in T3 and T4 $\rightarrow 2/5 = 40\%$ support

For example, consider the rule:

Milk $\rightarrow$ Bread

Confidence:

$$\text{Confidence}(\text{Milk} \rightarrow \text{Bread}) = \frac{\text{Support}(\text{Milk, Bread})}{\text{Support}(\text{Milk})}$$

$$= \frac{2}{4} = 50\%$$

How it improves sales

The store can use these relationships for:

1. Product placement – Place commonly purchased products near each other.
2. Combo offers – Create offers such as Milk + Bread.
3. Cross-selling – When a customer purchases Milk, recommend Bread.
4. Discounts – Provide discounts on related products.
5. Inventory management – Maintain sufficient stock of frequently associated products.
6. Online recommendations – E-commerce websites can recommend related products.

Practical example

If analysis shows that customers who purchase **Milk frequently purchase Bread**, the store can place Milk and Bread close together or offer a **Milk + Bread combo**.

Thus, association rule mining helps retailers understand customer buying behavior and increase sales through better product placement, recommendations, and promotional strategies.

Q2. Dataset: Transaction Data

TID	Items
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

Question:

Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

ANS:-

Apriori vs FP-Growth

The given dataset contains items A, B, and C.

Transactions:

- $T1 \rightarrow A, B$
- $T2 \rightarrow B, C$
- $T3 \rightarrow A, C$
- $T4 \rightarrow A, B, C$
- $T5 \rightarrow B, C$

Both Apriori and FP-Growth are frequent itemset mining algorithms.

Apriori

Apriori generates candidate itemsets and checks their support.

For example:

- 1-itemsets $\rightarrow A, B, C$
- 2-itemsets $\rightarrow AB, AC, BC$
- 3-itemset $\rightarrow ABC$

It repeatedly scans the database to determine which itemsets satisfy the minimum support.

FP-Growth

FP-Growth uses an FP-Tree (Frequent Pattern Tree). Instead of generating a large number of candidate itemsets, it compresses the transaction database into a tree structure and mines frequent patterns from it.

Comparison:-

Feature	Apriori	FP-Growth
Basic technique	Candidate generation	FP-Tree
Database scans	Multiple scans	Usually fewer scans
Candidate generation	Required	Not required
Memory	Can require more memory	Generally more efficient
Large datasets	Less suitable	More suitable
Small datasets	Simple and useful	Also works well
Implementation	Easier to understand	More complex

When to use

Apriori:

Use it when the dataset is relatively small and simplicity and easy understanding are important.

FP-Growth:

Use it for large transaction databases where candidate generation would become expensive.

Conclusion

Apriori is simple but may become slow because of repeated candidate generation and database scanning. FP-Growth is generally faster for large datasets because it avoids explicit candidate generation and uses an FP-Tree.

Q3. Dataset: Transaction Data

TID	Items
T1	A, B
T2	A
T3	B
T4	A, B
T5	A

Question:

Explain support and confidence using the dataset and describe their importance in association rule mining.

ANS:-

Support and Confidence

Support and confidence are two important measures used to evaluate association rules.

Given:

TID	Items
T1	A, B
T2	A
T3	B
T4	A, B
T5	A

There are 5 transactions.

1. Support

Support measures how frequently an item or itemset occurs in the entire dataset.

Support of A

A occurs in:

T1, T2, T4, T5 = 4 transactions

$$Support(A) = \frac{4}{5} \times 100 = 80\%$$

Support of B

B occurs in:

T1, T3, T4 = 3 transactions

$$Support(B) = \frac{3}{5} \times 100 = 60\%$$

Support of A and B

A and B occur together in T1 and T4 = 2 transactions

$$Support(A, B) = \frac{2}{5} \times 100 = 40\%$$

2. Confidence

Confidence measures how often B occurs when A occurs.

For the rule:

$A \rightarrow B$

$$\begin{aligned} Confidence(A \rightarrow B) &= \frac{Support(A, B)}{Support(A)} \\ &= \frac{2}{4} \times 100 = 50\% \end{aligned}$$

Therefore, there is a 50% confidence that a customer who buys A also buys B.

Importance:-

Support helps identify frequent and useful item combinations and removes very rare patterns.

Confidence measures the reliability of an association rule and helps determine whether the rule is useful for prediction or recommendation.

For example, a retailer could use a strong $A \rightarrow B$ rule to recommend B when a customer purchases A.

Conclusion

Support tells us how common a combination is, while confidence tells us how reliable the relationship is.

Q4 . Dataset: Multilevel Data

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Question:

Explain multilevel association rules using the dataset and compare them with multidimensional.

ANS:-

Multilevel Association Rules

Multilevel association rules discover relationships at different levels of abstraction in a product hierarchy.

Given:

- $T1 \rightarrow \text{Electronics, Mobile}$
- $T2 \rightarrow \text{Electronics, Laptop}$
- $T3 \rightarrow \text{Electronics, Mobile}$
- $T4 \rightarrow \text{Electronics, Mobile}$
- $T5 \rightarrow \text{Electronics, Laptop}$

Here, the hierarchy is:

Electronics $\rightarrow$ Mobile/Laptop

Level 1: General category

All 5 transactions contain Electronics.

$$\text{Support}(\text{Electronics}) = \frac{5}{5} \times 100 = 100\%$$

Level 2: Specific products

Mobile occurs in T1, T3, T4:

$$\text{Support}(\text{Mobile}) = \frac{3}{5} \times 100 = 60\%$$

Laptop occurs in T2 and T5:

$$\text{Support}(\text{Laptop}) = \frac{2}{5} \times 100 = 40\%$$

Therefore, rules can be discovered at different levels.

For example:

Electronics → Mobile

or more specifically:

Electronics products → Mobile

The purpose is to identify both broad and detailed customer purchasing patterns.

Multilevel vs Multidimensional Association Rules

Multilevel Association	Multidimensional Association
Uses different levels of a hierarchy	Uses multiple attributes/dimensions
Example: Electronics → Mobile	Example: Age → Product
Focuses on abstraction levels	Focuses on different properties
Uses product/category hierarchy	Uses attributes such as age, income, location
Useful for product analysis	Useful for customer behavior analysis

Example of multidimensional rule

A retailer may discover:

Young customers + High Income → Mobile

Here, Age and Income are different dimensions.

Conclusion

Multilevel association rules analyze relationships from general categories to specific products, whereas multidimensional association rules analyze relationships among different attributes or dimensions.

Q5. Dataset: Large Transaction Sample

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

Question:

Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

ANS:- Challenges in Mining Large Datasets

The given sample contains transactions involving A, B, and C, but in a real-world large transaction database there may be millions of transactions and thousands of different items.

The main challenges are:

1. Large dataset size

Large databases may contain millions of transactions. Storing and processing all data requires significant memory and computing power.

2. Large number of candidate itemsets

As the number of items increases, the number of possible item combinations grows rapidly.

For example:

- 1-itemsets $\rightarrow$ A, B, C
- 2-itemsets $\rightarrow$ AB, AC, BC
- 3-itemset $\rightarrow$ ABC

With thousands of items, this becomes extremely large.

3. Multiple database scans

Traditional algorithms such as Apriori may scan the database many times, increasing processing time.

4. Memory requirements

Large transaction databases and candidate itemsets can consume considerable memory.

5. Processing time

Finding frequent patterns in millions of transactions can take a long time if inefficient algorithms are used.

Scalable methods

Several techniques can address these challenges:

FP-Growth

FP-Growth uses an FP-Tree and avoids extensive candidate generation. It is suitable for large transaction datasets.

Data partitioning

The dataset can be divided into smaller partitions and processed separately. This reduces memory and processing requirements.

Sampling

A representative sample of the large dataset can be analyzed to obtain useful patterns more quickly.

Parallel and distributed processing

Large datasets can be processed across multiple machines using distributed computing technologies. This allows different parts of the data to be processed simultaneously.

Efficient data structures

Structures such as FP-Trees compress transaction information and reduce unnecessary processing.

Conclusion

Mining large datasets is challenging because of huge data volume, candidate explosion, memory usage, and processing time. Scalable approaches such as FP-Growth, partitioning, sampling, parallel processing, and distributed computing make association rule mining faster and more practical for large-scale applications.